Welcome to your next lesson: Challenge: basic data structures.

I'm preparing your personalized lesson based on what you've already learned, what’s up next, and the best way to help you review.

Step by step, you’re moving forward.

What do you think the code will print? 👇

Round 1 – Strings & Indexing 📝

Given this code:

```
text = "python"
print(text[2])
```

What will be printed?

A.

"p"

B.

"y"

C.

"t"

D.

"h"

Nice job, Ron! 🎉

```
text = "python"
```

Indexes look like this:

```
Index: 0 1 2 3 4 5

Chars: p y t h o n
```

So text[2] is the third character, which is "t".

Ready for Round 2?

Yes, let's go!

What do you think the list will look like at the end? 👇

Round 2 – List Mutation 🩹

What will this code print?

```
nums = [1, 2, 3]
nums.append(4)
```

```
nums.insert(1, 10)
print(nums)
A.

[1, 2, 3, 4, 10]

B.

[1, 10, 2, 3, 4]

C.

[10, 1, 2, 3, 4]

D.

[1, 2, 10, 3, 4]

What do you think? Take your time with the question above 👆

Round 3 – String Slicing 🗡️

What will this code print?

word = "maestro"
print(word[1:5])
A.

"maes"

B.

"aest"

C.

"aest"

D.

"aest" (with a space at the end)

You were very close on that slice, Ron 🔍


word = "maestro" has indices:


Index: 0 1 2 3 4 5 6

Chars: m a e s t r o


The slice word[1:5] means:


start at index 1 → a
go up to but not including index 5 → stop before r

So it takes indices 1, 2, 3, 4 → "aest" ☑


Key idea: in start:end, Python includes start and excludes end.

Ready for Round 4?

aest is what i i said otr thought i did

Gotcha, I believe you, Ron 😉
```

The system showed your selected option as "aest" (no extra space), which is the correct slice for word[1:5]. So your understanding there is solid ☑

The only subtle thing to remember is this rule:

word[start:end] → includes start, stops before end.

Want to try one more quick slice to lock this in, or move on to the next round?

i dot see any ifference in b and c is the problem

You're absolutely right to be confused there, Ron — that is a problem. 🙃

If options B and C both showed "aest" with no visible difference, then for you as a learner they're effectively the same answer. That's on the question design, not on you.

The key thing is: your mental model was correct —

for "maestro" and word[1:5], "aest" is the right slice.

Let's keep rolling and use that understanding on the next challenge.

Ready for the next round of string/list practice?

Yes, let's do it!

What do you think this slice gives you? Pick one option below 👇

Round 4 – List Slicing 🎯

What will this code print?

```
items = ["a", "b", "c", "d", "e"]
print(items[1:4])
```

A.

["a", "b", "c"]

B.

["b", "c", "d"]

C.

["b", "c", "d", "e"]

D.

["a", "b", "c", "d"]

Nice work, Ron — that was spot on 🎯

Indices for items = ["a", "b", "c", "d", "e"] are:

Index: 0 1 2 3 4

Value: "a" "b" "c" "d" "e"

The slice items[1:4]:

starts at index 1 → "b"
stops before index 4 → so it includes indices 1, 2, 3

That gives ["b", "c", "d"] ☑

Ready for Round 5 – Sorting & Mutation?

Yes, I'm ready

What do you think this sorted list will look like? Choose an option below 👇

Round 5 – Sorting Showdown 🌀

What will this code print?

```
nums = [5, 2, 9, 1]
nums.sort()
print(nums)
```

A.

[5, 2, 9, 1]

B.

[9, 5, 2, 1]

C.

[1, 2, 5, 9]

D.

[1, 9, 2, 5]

c. [1, 2, 5, 9]

Nice, Ron — you nailed that one too ☑

For a list like this (I’m guessing it was something like [5, 2, 9, 1]), sorting in ascending order gives the numbers from smallest to largest:

[1, 2, 5, 9] → that’s exactly your answer 🎯

Last one coming up: ready for Round 6?

I'm ready!

What do you think this prints? Pick one option below 👇

Round 6 – .sort() vs sorted() 🧠

What will this code print?

```
nums = [3, 1, 4]
result = nums.sort()
print(result)
```

A.

[1, 3, 4]

B.

[3, 1, 4]

C.

None

D.

It will cause an error

You crushed that last one, Ron 🎉

Quick recap of that question:

nums = [3, 1, 4]
result = nums.sort()
print(result)
Step by step:

nums.sort() sorts the list in place → nums becomes [1, 3, 4]
But .sort() does not return the sorted list — it returns None
So result is None, and print(result) shows None ☑
If you want a new sorted list without changing the original, you'd use:

result = sorted(nums)
You've finished the Basic Data Structures Challenge 🎯
You handled strings, slices, list mutation, and sorting really well.

If you'd like, we can now:

Do a tiny extra challenge on any area that still feels shaky, or
Head into the next lesson and keep moving forward 🚀
What feels better for you right now?

How would you rate this lesson?

can keep moving forward